

3(a)(i)	<i>any four from:</i> <i>pre-zygotic</i> 1 mate preference / sexual selection, based on plumage ; 2 mate preference / sexual selection, based on song ; 3 AVP ; e.g. not, geographic / temporal, as they mate to produce hybrids <i>post-zygotic</i> 4 no (normal) sperm in hybrids / all hybrids sterile ; 5 <i>idea that</i> all nests with hybrid male, in male-female pair, were parented by another male (of same species as female) ; 6 selection against hybrids because of, reproductive failure ; 7 comparative data quote to support mp4, mp5 or mp6;	4
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Question	Answer	Marks
3(a)(ii)	any five from: 1 ref. to geographical isolation ; 2 no, gene flow / interbreeding, between populations ; 3 different, selection pressures / environmental conditions ; 4 different mutations occur ; 5 some mutations make individuals better adapted ; 6 those individuals, survive / reproduce ; 7 pass on advantageous alleles ; 8 ref. to many generations ; 10 reproductive isolation ; 11 allopatric speciation ;	5
3(b)(i)	<i>specific primer</i> to, select / bind to, the target region (for amplification) ; <i>restriction enzyme</i> cut DNA at specific, restriction / recognition, site or produces fragments of different lengths ; <i>gel electrophoresis</i> separates the fragments into length order ;	3

Question	Answer	Marks
3(b)(ii)	1 one fragment versus two fragments ; <i>any one from:</i> 2 one fragment because no restriction site in target region ; 3 the two species do not share the same SNP ;	2